

June 5, 2026

Spotlight: The GLP-1 Manhattan Project: How Eli Lilly Is Using Today's Metabolic Cash Flow to Build Tomorrow's Biological Intelligence

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These numbers are shocking. It's like we got a new frontier AI model but for the body. – Bryan Johnson regarding Eli Lilly's recent retatrutide data

I want to make sure my feelings on the importance of this paper are clear with some extreme hot takes to start. First, I believe by the end of this decade, Eli Lilly could be the largest company in the US. Second, Eli Lilly is becoming the most important AI company in the world. Third, peptides this decade are equivalent to API keys from the prior decade. Hopefully that got your attention enough to read what is the longest paper I have written in a long time and this is after I cut it down by 50%. It took almost a year of research to gather the thoughts and information to support these hot takes.

If you go back and look at history, every major technology cycle begins with a breakthrough product, but the companies that define the cycle and the product rarely stop there. The breakthrough gives them revenue. Revenue gives them capital. Capital gives them the ability to build infrastructure. Infrastructure gives them the ability to compound. We are watching that now with the hyperscalers as they race to win the next technology cycle built on AI.

In my opinion, that is the story now unfolding at Eli Lilly.

Most investors understand Lilly and its recent success through the lens of GLP-1s. Mounjaro and Zepbound which have transformed the company from a successful pharmaceutical business into one of the most important growth companies in the world. The GLP-1 market has become a commercial phenomenon because it sits at the center of obesity, diabetes, metabolic dysfunction, cardiovascular risk, sleep apnea, inflammation, and potentially many other diseases tied to modern health. It hurt consumer staples companies and recently was impactful to Victoria's Secret's earnings and stock price.

Focusing only on Lilly's current revenue misses the larger strategic point.

GLP-1s are the funding engine for Lilly's next act.

That next act is artificial intelligence for human biology.

The most important thing happening at Lilly is the way cash flow from the GLP-1 franchise is giving the company the balance sheet, ambition, data and urgency to build a specialized AI infrastructure stack for drug discovery, genetic medicine, manufacturing, and eventually longevity. While the world is focused on general-purpose LLM companies racing to build ever-broader intelligence models looking for consumer and enterprise adoption while fighting commoditization risk and explosive rising costs for the infrastructure, Lilly is building something narrower and potentially far more defensible: a proprietary biological intelligence system trained on decades of molecular, preclinical, clinical, manufacturing, and patient knowledge.

General AI vs. Biological AI

The general AI race is horizontal. OpenAI, Google, Anthropic, Meta, xAI and China are competing to build models that can reason across everything: software, writing, coding, math, images, video, research, enterprise work, consumer interfaces, and eventually autonomous agents. It is a massive race, and it is also a crowded race that is deflationary on one side and

costly on the buildout. The same basic ingredients are available to many competitors: capital, GPUs, talent, synthetic data, public data, enterprise data, and distribution.

Lilly's race is different. It is vertical. It is specialized. It is domain-specific. It is built around biology, chemistry, medicine, and the industrial process of turning scientific knowledge into approved therapies. This model is designed to ask better biological questions, design better molecules, run better experiments, predict safety and efficacy earlier, manufacture more efficiently, and shorten the time between hypothesis and medicine.

That is a fundamentally different business model than building a chatbot that turns into agents.

It is closer to building a new operating system for the human body.

The reason GLP-1 revenue is so important is that this type of infrastructure cannot be built cheaply. It requires compute, data, robotics, laboratories, scientists, engineers, partnerships, acquisitions, regulatory knowledge, manufacturing scale, and patience. Many AI-native biotech companies have exciting models, but they do not have Lilly's cash flow. Many technology companies have compute, but they do not have Lilly's proprietary biological data. Many pharmaceutical companies have data, but they may not have Lilly's current growth engine or urgency.

Lilly sits at a unique intersection: massive revenue growth today, massive disease markets tomorrow which currently make up close to 20% of consumer expenditures, and a strategic need to convert current dominance into long-term infrastructure.

That is why the Manhattan Project analogy works.

The Manhattan Project was a concentrated mobilization of capital, talent, physics, engineering, manufacturing, and national urgency around one problem. Lilly's AI effort has a very different purpose, but the structure of the analogy is useful. This is an effort to assemble the pieces of a new scientific production system. Data, compute, AI agents, wet labs, dry labs, robotics, genetic medicine, and manufacturing are being pulled into one loop. The goal is to make discovery faster, cheaper, more predictive, and more continuous.

From Drug Discovery to a Scientific Production System

The old drug discovery model was slow because biology is complex and the experimental cycle is painful. A scientist forms a hypothesis. A molecule is designed. Experiments are run. Data comes back. Teams analyze it. New hypotheses are formed. The process repeats, often for years. Most compounds fail. Many failures occur late, after enormous cost and time have already been spent. The industry has always lived with this attrition because biology is hard.

AI changes the bottleneck.

The first phase of AI in drug discovery was mostly about prediction: predicting protein structures, molecular properties, toxicity, binding affinity, drug-like characteristics, and disease pathways. That was useful, but still limited. The more important phase is agentic AI. An AI agent can read literature, form hypotheses, choose tools, design experiments, interact with robotic labs, analyze results, write reports, and begin the next cycle.

In other words, AI agents can help automate the Design-Make-Test-Analyze loop that sits at the heart of pharmaceutical research.

This is where Lilly's strategy becomes more powerful. A general LLM agent may help with coding, emails, research, or office workflows. A Lilly agent connected to proprietary molecular data, safety data, preclinical data, wet labs, robotics, and manufacturing workflows becomes part of the scientific process itself. The agent participates in the creation of new biological knowledge.

That is a profound difference.

Failure Becomes Training Data

Lilly's data is the real moat in biological AI. The internet created a massive training set for general models. Biology does not have the same open training set. The highest-value data sits inside pharmaceutical companies, academic labs, clinical trials, failed experiments, manufacturing systems, toxicology studies, and patient records. Much of it is messy, expensive, proprietary, and difficult to interpret.

You cannot scrape your way into Lilly's history of molecules. You cannot easily recreate decades of experimental failure.

And in drug discovery, failure is training data.

That is one of the most underappreciated ideas in the AI-pharma story. The pharmaceutical industry has spent decades generating enormous amounts of negative information. Compounds that did not work. Targets that failed. Doses that were unsafe. Pathways that looked promising but did not translate. Trials that disappointed. Historically, much of that failure was viewed as cost. In an AI world, it becomes an asset.

A model that knows what failed, why it failed, and under what conditions it failed has a better chance of avoiding the same mistakes. The graveyard of failed drug discovery may become one of the most valuable datasets in the world.

This is also why Lilly's strategy is more defensible than the general LLM race. In the LLM world, many companies can train on similar public data, license similar datasets, generate synthetic data, and chase similar benchmarks. In pharma, the most important benchmark is whether the model can help create a molecule that works in the human body, avoids unacceptable toxicity, can be manufactured, can pass regulatory review, and can become a medicine.

That requires a closed loop between model and reality.

The lab is the judge. The patient is the ultimate judge.

TuneLab: The Distributed Learning Network

However, most of what I have written so far is from a top-down perspective. The part that has gathered my focus is the same as the ones for memory, optical, power, the rack, edge and advanced semiconductor packaging. The partnerships and actual announcements happening right now. AI moves at an exponential pace and so the companies also have to connect at an exponential pace with the ecosystem and this has been the driving factor for Eli Lilly starting with TuneLab.

Lilly's TuneLab is one of the clearest signs of this direction. On the surface, TuneLab looks like a generous platform: Lilly gives biotechnology companies access to AI-enabled drug discovery models trained on decades of Lilly research data and built through more than \$1 billion of research investment. Strategically, TuneLab is much more than a tool. It is a give-to-get strategy. Lilly is trading access to its AI models for something potentially more valuable over time: participation in a broader biological learning network.

The key concept is federated learning. In a traditional data model, companies would have to pool their data in one central location. That is nearly impossible in biotech because the most valuable data is proprietary, sensitive, regulated, and competitively guarded. Federated learning changes the structure. The model can learn across distributed datasets while the raw data remains inside each participant's environment. The partner keeps control of its data, while the shared model improves from the patterns, outcomes, and experimental feedback generated across the network.

That matters because the bottleneck in biological AI is data diversity as much as compute. A single pharmaceutical company, even one as large as Lilly, cannot internally explore every chemical space, every target, every disease pathway, every modality, and every experimental failure. A network of biotechs can. Each partner may be working on a different target, a different mechanism, a different molecule class, or a different therapeutic area. Each one is generating data Lilly would never have produced internally.

TuneLab creates a structure where those distributed efforts can improve the intelligence of the shared system.

This is the data flywheel. Lilly's historical data makes the initial models useful. That usefulness attracts biotech partners. Those partners apply the models to their own proprietary programs. Their experiments generate new learning. The model improves. A better model attracts more users. More users generate more diverse learning. The system compounds.

That is why TuneLab should be viewed as an outsourced R&D network. TuneLab sits inside Lilly Catalyze360, the company's broader venture, partnering, and incubation platform. That gives Lilly a front-row seat to emerging founders, targets, modalities, and therapeutic strategies before they are obvious to the public market. It can build relationships with companies before they reach major clinical milestones. It can see where scientific energy is moving. It can identify which approaches are gaining traction. And it can create a natural funnel for future licensing deals, joint ventures, partnerships, or acquisitions. Think Google X.

The Google X comparison is useful, but only partially. Like Alphabet's moonshot model, Lilly is creating a sandbox where high-risk, early-stage innovation can form around its infrastructure. Google X was largely designed to find Alphabet's next business beyond search and advertising. TuneLab is designed to reinforce Lilly's existing core business.

That makes TuneLab less like a moonshot factory and more like a biological data flywheel. Lilly is using its scale to become part of the platform layer for early-stage drug discovery. If enough biotechs use Lilly's models to screen compounds, predict toxicity, evaluate pharmacokinetics, and prioritize candidates, then Lilly's parameters begin to influence what the ecosystem considers a promising molecule.

In the broader AI arms race, algorithms will eventually commoditize. Proprietary data, specialized workflows, compute infrastructure, and feedback loops are the constraints. TuneLab is Lilly's attempt to own more of that loop. It turns outside innovation into inside learning. It turns smaller biotech companies into distributed nodes in a larger biological intelligence system. And it gives Lilly a structural advantage that can scale beyond the limits of its own R&D headcount.

The clean investor line is this: TuneLab is Lilly's attempt to turn the biotech ecosystem into a distributed learning network or a way to make outside innovation improve Lilly's inside intelligence.

External AI Partnerships: Buying Speed, Optionality, and Frontier Methods

TuneLab shows that Lilly wants to turn the broader biotech ecosystem into a distributed learning network. Lilly is also building relationships with some of the most important AI-native drug discovery companies in the world.

That matters because the winner in biological AI may be the company that knows which external tools to integrate, which platforms to partner with, and which discoveries to pull into its own development and commercialization machine.

The partnerships with Isomorphic Labs and Insilico Medicine are important signals.

Isomorphic Labs is especially meaningful because it comes from the DeepMind and AlphaFold lineage. AlphaFold changed the world's understanding of protein structure prediction. Isomorphic is trying to apply that same AI-native mindset to drug design. Lilly's collaboration with Isomorphic focuses on discovering small-molecule therapeutics against undisclosed targets. Strategically, that gives Lilly exposure to one of the most advanced AI drug-design engines in the world without needing to own all of that capability internally.

Insilico Medicine represents a different but equally important layer. Insilico has built a full-stack AI drug discovery platform spanning target discovery, molecule generation, and clinical-development prediction. Lilly's expanded relationship with Insilico suggests that the company is willing to use external AI engines as sources of actual drug candidates. That is an important shift. Lilly is using AI-native companies as upstream engines for future pipeline formation.

This is the right strategy for a pharmaceutical company with Lilly's scale. In general AI, the frontier model companies often try to own the entire stack because the model itself is the product. In specialized biological AI, the medicine is the final product. Lilly does not need to build every model, every interface, or every discovery engine itself. It needs to control the highest-value parts of the loop: proprietary data, capital allocation, clinical development, regulatory execution, manufacturing, distribution, and commercialization.

That is why these partnerships should be viewed as ROIC accelerators. Isomorphic gives Lilly access to frontier structural biology and small-molecule design. Insilico gives Lilly access to generative chemistry, target discovery, and AI-created programs. TuneLab gives Lilly a distributed network of biotech learning. NVIDIA gives Lilly the compute and physical AI infrastructure. Lilly's own data gives the system its proprietary memory.

Together, these pieces turn Lilly into an aggregator of biological intelligence.

The broader point is simple: Lilly is building a portfolio of intelligence sources. Some are internal. Some are partnered. Some are ecosystem-based. Some are embedded in external AI-native companies. This lowers the risk that Lilly misses the winning method while increasing the chance that an important breakthrough in AI drug discovery can eventually flow into Lilly's development machine.

That is the strategic power of incumbency in the AI-pharma era. The AI-native companies may have the models. Lilly has the cash flow, clinical infrastructure, regulatory knowledge, manufacturing scale, and commercial reach. By partnering early, Lilly can turn external AI innovation into internal pipeline optionality.

Specialized AI Is an ROIC Story

There is another important difference between general AI and specialized AI.

The general AI companies are mostly selling a labor story. Their models are designed to replace, augment, or compress human work across coding, customer service, research, legal workflows, sales, marketing, and back-office operations. The promise is a world of digital employees.

Lilly's specialized AI story is different. It is focused on improving the company's own return on invested capital.

In pharma, the largest cost is failed capital: molecules that do not work, trials that end too late, manufacturing processes that take too long to optimize, and biological hypotheses that consume years before reality proves them wrong. A specialized biological AI model is valuable because it can help Lilly ask better questions, kill weak programs earlier, design better molecules, run tighter experiments, and recycle every success and failure back into the system.

That is ROIC.

This also helps explain why secrecy may become more valuable. Lilly's pipeline page now states that Phase 1 projects are no longer disclosed. That change should not be ignored. Historically, early-stage pipeline disclosure helped investors understand the breadth and direction of a pharmaceutical company's research engine. In an AI-enabled world, early disclosure may carry a higher competitive cost.

A Phase 1 asset can reveal a target, a mechanism, a biological pathway, a modality, a disease strategy, or a combination logic that competitors can study with their own models. If AI reduces the time it takes rivals to learn from public information, then public information becomes more dangerous. Lilly's reduced disclosure therefore looks like a rational response to a world where biological intelligence is becoming faster, more computational, and more competitive.

The message is subtle but important: the pipeline itself is becoming proprietary infrastructure.

LillyPod: The Infrastructure of Biological Intelligence

There has been one way to make money since the launch of ChatGPT and that is invest in Nvidia and its partners and that's where this story goes. Lilly's NVIDIA infrastructure should be viewed not as IT, but as a new class of scientific instrument. At the heart of this strategy is LillyPod, an NVIDIA DGX SuperPOD powered by over 1,000 NVIDIA Blackwell Ultra GPUs. You have heard about AI factories. Well this is a big one. This is not a general-purpose cloud rental; it is a wholly owned, industrial-scale AI factory designed to manage the entire lifecycle of biological discovery from data ingestion and massive-scale molecular simulation to high-volume inference.

The shift here is from "artisanal" discovery to "industrialized" biology. While a productive wet-lab team might test roughly 2,000 molecular hypotheses per target, per year, LillyPod allows researchers to simulate and evaluate billions of hypotheses in parallel. By integrating this hardware with the NVIDIA BioNeMo platform and a proprietary data stack built on decades of experimental success and failure, Lilly is creating a closed-loop system where compute generates hypotheses, robotics run the experiments, and the resulting data continuously improves the models.

This infrastructure is the “bridge” between the world of bits and atoms: it enables the company to compress the biological bottleneck by turning the scientific process into a continuous, compounding computational feedback loop.

This is pharmaceutical infrastructure becoming AI-native.

The NVIDIA co-innovation lab extends this even further. The strategic direction is to connect AI to physical experimentation, robotics, manufacturing, and digital twins. That is the key step. In software, AI can improve quickly because code can be written, tested, and deployed almost instantly. Biology has always been slower because experiments require the physical world. Cells must grow. Assays must run. Animals or organoids may be studied. Manufacturing processes must be validated. Clinical trials take time.

The promise of agentic wet labs and robotic experimentation is that the cycle time compresses dramatically.

This is where the “atoms to bits” idea becomes central. AI is exploding in the world of bits. Biology is still anchored in atoms. The companies that win will be the ones that connect the two. Lilly is using GLP-1 cash flow to build the bridge between digital intelligence and physical biology. Compute generates hypotheses. AI agents design experiments. Robotic labs run them. Data returns to the model. The model improves. The next experiment begins.

Over time, that feedback loop becomes the asset.

Longevity Is a Network Problem

The longevity connection emerges naturally from this. Longevity is the interaction of many systems: metabolism, cardiovascular function, inflammation, immune decline, neurodegeneration, muscle loss, mitochondrial function, cellular senescence, genomic instability, and hormonal regulation. Traditional medicine has been organized around diseases. Longevity forces medicine to think in networks. That makes it an ideal AI problem.

Lilly’s current GLP-1 franchise already touches one of the most important longevity pathways: metabolic health. Obesity and diabetes increase the risk of cardiovascular disease, kidney disease, sleep apnea, liver disease, inflammation, and many other health problems. Improving metabolic function may therefore have effects across healthspan, beyond weight loss.

This is why the GLP-1 story is bigger than obesity.

It is about the possibility that one therapeutic class can improve multiple downstream systems tied to aging and chronic disease.

GLP-1s may only be the beginning. If Lilly can use the revenue from metabolic disease to build an AI engine for broader biology, the company can expand from treating chronic conditions to understanding the architecture of healthspan itself. That does not mean immortality. It means identifying the biological bottlenecks that cause people to lose function, develop disease, and age poorly. It means discovering interventions earlier. It means designing molecules faster. It means combining therapies in more intelligent ways. It means moving from population averages toward more personalized medicine.

The gene-editing angle strengthens this thesis. Lilly’s move into companies and partnerships focused on programmable genetic medicine suggests that the company understands the next phase of medicine may extend beyond chronic treatment. Some diseases may eventually be addressed through durable biological repair. A pill or injection taken repeatedly manages a pathway. A gene-editing therapy may alter the underlying biology more permanently. AI-designed tools for gene editing could become one of the most important areas of medicine because they expand what is programmable inside the body.

In this sense, the future of pharma starts to look less like traditional drug development and more like software development for biology.

The body is not software, and the analogy should not be pushed too far. Biology is messier, more adaptive, and more dangerous than code. But the direction is clear. Medicine is moving from broad chemical intervention toward increasingly

precise biological programming. GLP-1s modulate powerful metabolic pathways. Gene editors rewrite instructions. AI agents help design the tools. Robotics accelerates experimentation. Digital twins simulate systems. Data compounds. The entire process becomes more computational.

Peptides as API Keys for Human Software

This is where the comparison to the Mag 7 becomes useful.

The Mag 7 became valuable because they controlled the access layers of digital software. They owned the keys into search, mobile operating systems, cloud infrastructure, commerce, identity, social graphs, enterprise workflows, and accelerated compute. They controlled the points of access through which companies, consumers, developers, advertisers, and enterprises interacted with the digital world.

Peptides are similar, but in biology.

They are access keys into human software.

The body is not software in the literal computer-code sense, but it behaves like a layered information system. DNA stores instructions. RNA transmits messages. Proteins execute functions. Hormones coordinate signals. Receptors authenticate those signals. Feedback loops regulate the entire system. A peptide fits into that system as a small programmable sequence that can unlock a specific biological response.

In the digital world, an API key gives a developer permissioned access to a platform. In the biological world, a peptide, hormone, or ligand gives a molecule access to a pathway. The software platform is the human biological system. The codebase is DNA and cellular machinery. The authentication layer is receptor binding. The server response is the physiological response. The permission structure is pathway specificity. The platform owner is the company with the data, intellectual property, delivery mechanisms, manufacturing scale, and regulatory expertise to turn that biological access into medicine.

GLP-1s are the perfect example.

A GLP-1 drug is essentially a biological API key. It plugs into existing human metabolic software and changes the behavior of the system. It tells the body to feel fuller, slow gastric emptying, improve insulin response, reduce food noise, alter energy balance, and potentially influence inflammation, cardiovascular risk, kidney disease, liver fat, addiction pathways, and neurobiology.

That is why GLP-1s are bigger than obesity drugs.

They reveal that human software is programmable.

For the Mag 7, the market eventually learned that whoever controls the API layer controls the economics of the digital world. For Lilly, Novo Nordisk, and the next generation of biotech platforms, the market may be learning something similar: whoever controls the biological API layer may control the economics of future medicine.

The difference is depth.

The Mag 7 controlled access to external software: apps, devices, clouds, ads, payments, communication, and digital workflows. Peptides control access to internal software: metabolism, appetite, inflammation, immunity, fertility, muscle, aging, cognition, and cellular repair.

That makes the next platform shift potentially much larger. The first internet digitized information. The second wave digitized commerce and communication. AI is digitizing labor and intelligence. Peptides, gene editing, and AI-designed biology are beginning to digitize the human operating system itself.

If the Mag 7 became the API keys to the internet, peptides may become the API keys to the body. GLP-1s are the first mass-market proof that human software can be accessed, modulated, and monetized at platform scale.

That is the investor version of the Lilly thesis. The next great platform companies may control biological APIs, the keys that let AI, data, and engineered molecules interface directly with human metabolism, disease, aging, and performance.

M&A as Buying More Biological API Keys

If peptides are API keys into human software, then Lilly's M&A strategy should be viewed as an effort to buy more keys, more pathways, more modalities, and more optionality.

On the surface, Lilly's dealmaking looks like traditional pharma pipeline building. Large pharmaceutical companies have always acquired assets, bought programs, licensed molecules, and partnered with smaller companies to replace future revenue. But in Lilly's case, the pace and nature of the activity suggest something larger. The company is using GLP-1 cash flow to buy speed, biological data, modality exposure, and strategic control across the next era of medicine.

This matters because AI does not eliminate the need for external innovation. It increases the value of knowing which external innovation to buy.

If biological AI improves Lilly's ability to evaluate targets, mechanisms, molecules, and modalities, then M&A becomes more powerful. The company is no longer simply buying assets because they look promising in isolation. It can use its internal data, AI models, clinical knowledge, and manufacturing expertise to decide which assets fit into a much larger biological intelligence system. That turns M&A into a form of capital allocation enhanced by AI.

This is another reason the GLP-1 cash flow is so important. Lilly has the financial flexibility to act while the biological AI landscape is still forming. It can buy gene-editing platforms, RNA-editing technologies, oncology programs, oral therapeutics, inflammatory disease assets, vaccine developers, and AI-native discovery programs. Some of these deals will become products. Some will become data. Some will become scientific knowledge. Some will become manufacturing or modality capabilities. All of them can feed the machine.

That is the key point: in an AI-enabled pharma model, M&A is not only about buying revenue. It is about buying learning.

Verve gives Lilly exposure to one-time genetic treatments for cardiovascular disease. Ascidian gives Lilly access to RNA exon-editing technology for rare inherited kidney diseases. Scorpion expands the oncology pipeline with a targeted cancer program. Morphic adds oral integrin biology in inflammatory bowel disease. Insilico gives Lilly access to AI-native discovery programs. Isomorphic provides exposure to frontier AI drug-design intelligence. Each deal adds a different type of biological option.

Together, these moves show that Lilly is building across modalities, not just molecules.

That is important because the future of medicine may not be dominated by one therapeutic approach. It may be a portfolio of programmable tools: peptides, antibodies, small molecules, RNA therapies, gene editors, cell therapies, radiopharmaceuticals, and AI-designed combinations. Lilly's M&A strategy gives it more shots on goal across that expanding biological toolkit.

This is also where the ROIC argument becomes stronger. A company with a weak discovery engine can destroy value through M&A by overpaying for assets it does not understand. A company with a strong biological intelligence engine may be able to do the opposite. It can identify better targets earlier, understand fit more quickly, integrate assets more effectively, and kill weaker programs before they consume too much capital.

That is why Lilly's dealmaking should not be separated from LillyPod, TuneLab, Isomorphic, Insilico, and its internal data moat. They are all part of the same strategy. Lilly is building the infrastructure to discover internally, learn externally, partner selectively, acquire aggressively, and then push the best assets through one of the strongest clinical, regulatory, manufacturing, and commercial machines in healthcare.

This is what makes the company so different from a small AI biotech. An AI-native startup may have a brilliant model, but it usually lacks the capital, clinical infrastructure, manufacturing scale, regulatory experience, and global distribution needed to turn a discovery into a medicine. Lilly has those assets already. The more intelligence it adds upstream, the more valuable its downstream machine becomes.

M&A is therefore confirmation of the thesis.

Lilly is using today's metabolic cash flow to buy tomorrow's biological optionality. It is turning GLP-1 profits into a broader portfolio of AI, data, modalities, platforms, and therapeutic shots on goal. In a world where biology is becoming more computational, that may be one of the highest-return uses of capital available.

The Consumer Is Learning the Language of Biology

There is another reason this analogy matters: consumers are beginning to understand it before investors fully do.

Search interest for "peptides" has surged as GLP-1s moved from a diabetes and obesity category into mainstream culture. The word itself has crossed over from laboratories and clinical journals into TikTok, skincare, wellness clinics, Reddit threads, celebrity weight-loss discussions, and anti-aging conversations. That is important because every major platform shift requires a consumer translation layer. The Mag 7 did not become dominant only because developers understood APIs. They became dominant because consumers adopted the products those APIs enabled: search, smartphones, social graphs, cloud software, digital commerce, streaming, and now AI assistants. The technical layer became economically powerful once the consumer layer made it visible.

Peptides may be entering a similar moment in biology.

GLP-1s have educated the public that a small biological signal can profoundly change appetite, metabolism, body composition, and health outcomes. That awareness has created a halo around the broader peptide category. Some of it is legitimate medicine. Some of it is skincare. Some of it is biohacking. Some of it is a regulatory gray area. But the cultural signal is clear: consumers are starting to believe that biology is programmable.

That is the confirmation investors should pay attention to.

The consumer is no longer only asking for a drug to treat a disease. The consumer is asking for access to a pathway: weight loss, recovery, muscle, sleep, skin quality, inflammation, fertility, cognition, and aging. In the digital era, the killer app was access to information, commerce, communication, and software. In the biological era, the killer app may be access to human performance, metabolic control, and healthspan.

This does not mean every peptide trend is safe, proven, or investable. In fact, the gray-market boom around unapproved research peptides may be a warning that demand is moving faster than regulation and clinical evidence. But that is also part of the signal. Consumer curiosity is outrunning the existing medical system. Whenever that happens at scale, a new platform layer usually forms. The winners are rarely the unregulated edge players. They are the companies that can take consumer demand, validate it clinically, protect it with intellectual property, manufacture it reliably, distribute it safely, and wrap it inside data and AI feedback loops.

That is why Lilly's position is so important. GLP-1s are generating revenue and teaching the market that peptides can be mass-market biological interfaces. The consumer is learning the language of human software. Lilly is building the infrastructure to industrialize it.

Competition Exists, but the Race Is Different

The paper I wrote last year was a broader one on pharma and Lilly absolutely has competitors. Novo Nordisk is a major force in metabolic disease. Roche, Novartis, AstraZeneca, Sanofi, Merck, Pfizer, and many others are investing in AI. Recursion, Isomorphic, Insilico, Xaira, and other AI-native companies are building important platforms. NVIDIA will partner broadly. The field will be competitive.

But Lilly's competition is not the same as the LLM race.

It is not a clean benchmark race where everyone trains broadly similar models and competes for the same general intelligence layer. Lilly's advantage is that its model is specialized, its data is proprietary, its workflows are internal, and its capital comes from a massive operating business. Competitors can buy GPUs. They cannot easily buy Lilly's historical data. They cannot instantly recreate its therapeutic focus, manufacturing scale, clinical infrastructure, regulatory

knowledge, or GLP-1 cash flow. They can build models, but they may not be able to build the same closed-loop biological system.

This is where the investment story becomes powerful.

GLP-1 revenue is strategic fuel. Lilly can reinvest today's profits into the infrastructure that may create tomorrow's medicines. Each new molecule, each failed experiment, each clinical trial, each manufacturing improvement, and each partner interaction can become more data for the system.

The result is a compounding loop: GLP-1s generate cash; cash funds AI infrastructure; AI infrastructure improves discovery; improved discovery expands the pipeline; the pipeline creates more data and revenue; the data improves the models.

That is how a pharmaceutical company becomes an AI infrastructure company without investors fully noticing.

The market often struggles with this kind of transition because it tries to fit companies into old categories. Lilly is a pharma company. NVIDIA is an AI chip company. OpenAI is a model company. But the future does not respect those categories. NVIDIA is moving from chips to AI factories. OpenAI is moving from chat to agents. Lilly is moving from drugs to biological intelligence.

The convergence is the story.

Conclusion: From Metabolic Cash Flow to Biological Intelligence

For Lilly, the GLP-1 franchise may be remembered in two ways. The first is obvious: as one of the largest therapeutic breakthroughs in modern medicine. The second may prove even more important: as the financial bridge that allowed Lilly to build a new kind of drug discovery machine.

Mounjaro and Zepbound are treating metabolic disease today, but the cash flow they generate may fund the AI systems that discover future therapies for cardiovascular disease, neurodegeneration, cancer, immune dysfunction, genetic disease, and aging-related decline.

That is the deeper meaning of the Lilly story.

This is about the possibility that the largest healthcare revenue engine of this decade is financing a specialized AI platform for the next era of medicine.

The LLM companies are building general intelligence for the world of information. Lilly is building specialized intelligence for the world of biology. The first race may change how we work. The second may change how we age.

That is why Lilly deserves to be viewed as a GLP-1 leader and one of the most important AI infrastructure companies in healthcare. Its advantage is that the capital is arriving at the precise moment when biology is becoming computational, agents are becoming scientific workers, consumers are learning that biological pathways can be accessed, and proprietary data is becoming the new source of pharmaceutical power.

GLP-1s are funding the machine. AI is training the machine. Lilly's data is differentiating the machine. TuneLab is expanding the machine. Isomorphic and Insilico are extending the machine. Consumer demand is validating the machine. And the long-term output may be a new industrial model for discovering, designing, manufacturing, and personalizing medicine.

If the first phase of AI was about language, the next phase will be about life.

Lilly is positioning itself for that transition. The company's GLP-1 franchise gave it the cash flow. NVIDIA gives it the compute. TuneLab gives it the distributed learning network. Isomorphic gives it exposure to frontier drug-design intelligence. Insilico gives it access to AI-native discovery programs. Agentic wet labs give it the feedback loop. Genetic

medicine gives it the programmable frontier. Longevity gives it the ultimate market. And peptides give it the language of access between artificial intelligence and the human operating system.

That is the thesis: Lilly is using the profits from the metabolic disease revolution to finance a Manhattan Project for biological intelligence.

And if that system works, the company will be building the infrastructure through which future medicines are discovered.

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